Bonded anchor pryout test models

Assignment: Models 1–3 — updated edition

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Updated edition. This supersedes the July 2026 issue. The fixture plate is now $100\times400\times20$ mm (it was $80 \times 80 \times 20$ mm) and the embedment depth is $h_{\mathrm{eff}} = 70$ mm (it was 80 mm).

Each team shall construct, run and document all three model setups. Apply the common requirements below to Models 1–3 and apply each model-specific requirement in its corresponding section.

Common model-setup requirements

Material properties

The concrete properties are given in Table 1, i.e. grade C20/25. In the Table: E_{c} denotes the Young's modulus, ν_{c} denotes the Poisson's ratio, f_{cy} the yield strength, f_{cu} the ultimate strength, f_{bu} the biaxial strength, f_{tu} the ultimate tensile strength, $G^{\mathrm{f}}_{\mathrm{Ic}}$ the fracture energy in mode I fracture and ρ_{c} the density of concrete.

The anchor and the steel fixture components are made from structural steel with known elastic properties and density, see Table 2. In the Table: E_{s} denotes the Young's modulus, ν_{s} the Poisson's ratio and ρ_{s} the density of steel. The same steel definition is used for every steel region of the model, and each region is a separate material and element set.

Mortar properties are given in Table 2. The mortar is elastic, with different stiffness in the normal and the tangential directions, $E_{\mathrm{nm}}, G_{\mathrm{nt}}$; the values entered in each deck are listed with the cohesive section in the corresponding example finite element model.

Parameter	Value	Units
E_{c}	3.03×10^4	MPa
ν_{c}	0.2	–
f_{cy}	9.33	MPa
f_{cu}	28	MPa
f_{bu}	32.48	MPa
f_{tu}	2.2	MPa
$G^{\mathrm{f}}_{\mathrm{Ic}}$	133.0×10^{-3}	N/mm
ρ_{c}	2.4×10^{-6}	kg/mm^3

Table 1: Concrete properties.

Parameter	Value	Units
E_{s}	2×10^5	MPa
ν_{s}	0.33	–
ρ_{s}	7.85×10^{-6}	kg/mm^3
E_{m}	5400	MPa
ρ_{m}	2×10^{-6}	kg/mm^3

Table 2: Steel and mortar properties.

Boundary conditions

The concrete boundary conditions in Table 3 apply to all three models. The model is cut on the symmetry plane $z = 0$. Apply $u_3 = 0$ to the concrete, steel and mortar nodes on that plane. The model-specific symmetry sets and loading conditions are given with each example finite element model.

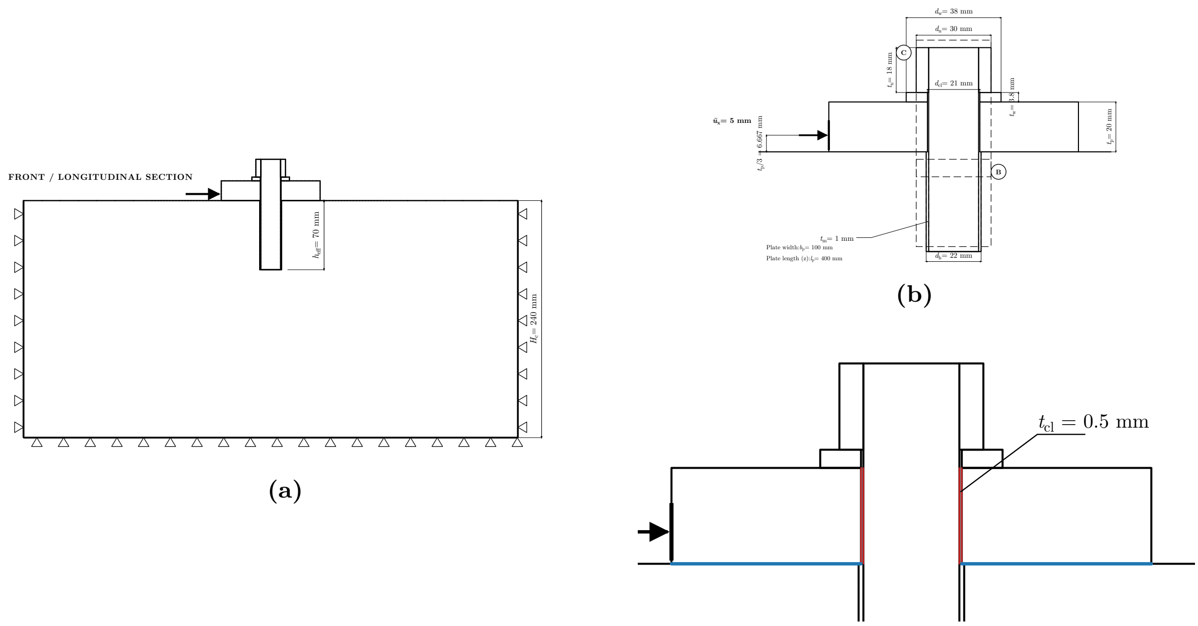
Node set	Location	DOFs	Condition
`concrete-1.left`	$x = -250$	1,2,3	$u_1 = u_2 = u_3 = 0$
`concrete-1.right`	$x = +250$	1,2,3	$u_1 = u_2 = u_3 = 0$
`concrete-1.back`	$z = -250$	1,2,3	$u_1 = u_2 = u_3 = 0$
`concrete-1.bottom`	$y = -240$	1,2,3	$u_1 = u_2 = u_3 = 0$
`concrete-1.z_symm`	$z = 0$	3	`zSYMM`: $u_3 = 0$

Table 3: Boundary conditions common to Models 1–3.

1 Model 1: Monolithic plate–anchor–washer–nut with self-contact

1.1 Model setup

This document describes the **explicit-fixture** variant of the model family. The anchor passes through a clearance hole in a rectangular steel plate and is held down by a flat washer and a hexagonal nut, both of which are meshed as separate bodies.



(a) (b) (c)

Figure 1: Bonded anchor model with the explicit washer–nut fixture. a Section on the symmetry plane $z = 0$, together with the boundary conditions of the concrete slab; the horizontal displacement u_{x} is introduced on the negative-x edge of the steel plate, at the height $t_{\mathrm{p}}/3 = 6.667$ mm above the concrete surface. b Detail near the anchor: mortar thickness t_{m}, borehole diameter $d_{\mathrm{h}} = d_{\mathrm{a}} + 2t_{\mathrm{m}}$, plate length b_{p}, washer thickness t_{w} and nut height t_{n}. c Clearance and contact definitions of the fixture: the radial clearance $t_{\mathrm{cl}} = 0.5$ mm between the shaft and the wall of the plate hole, the shaft/plate-hole interface carrying self contact inside the monolithic steel part (red), and the plate underside in contact with the concrete surface (blue).

1.1.1 Geometry

The model consists of a cylindrical steel anchor with diameter $d_{\mathrm{a}} = 20$ mm, embedded till depth $h_{\mathrm{eff}} = 70$ mm, in an orthogonal parallelepiped concrete slab with dimensions (length: $W_{\mathrm{c}} = 500$ mm, width: $W_{\mathrm{c}} = 500$ mm, height: $H_{\mathrm{c}} = 240$ mm $> 3h_{\mathrm{eff}}$). A cylindrical hole is present inside the concrete slab with diameter $d_{\mathrm{h}} = d_{\mathrm{a}} + 2t_{\mathrm{m}} = 22$ mm, where $t_{\mathrm{m}} = 1$ mm is the mortar thickness. The anchor is placed inside the hole so that their axes along the height match. The mortar region is modeled as a hollow cylinder sleeve with height h_{eff} and constant thickness

t_m between the anchor and the surrounding concrete.

The anchor is connected to a steel plate on top with dimensions (extent along x: $b_p = 100$ mm, extent along z: $l_p = 400$ mm, height: $t_p = 20$ mm). The plate contains a hole in the middle, through its thickness, with the clearance diameter $d_{cl} = d_a + 1 = 21$ mm, i.e. a radial clearance $t_{cl} = (d_{cl} - d_a)/2 = 0.5$ mm between the shaft and the hole wall. The anchor is connected to the plate through the fastener: the shaft is extended above the concrete slab by $t_u = t_p + t_w + t_n = 41.8$ mm, a flat washer of outer diameter $d_w = 1.9d_a = 38$ mm, bore $d_{cl} = 21$ mm and thickness $t_w = 0.19d_a = 3.8$ mm is placed on the plate top, and a hexagonal nut of width across flats $d_n = 1.5d_a = 30$ mm (across corners 34.64 mm) and height $t_n = 0.9d_a = 18$ mm is placed on the washer. The washer and nut proportions follow DIN 125A / ISO 7089 and ISO 4032 / DIN 934 respectively, as plain prisms of revolution and a plain hexagonal prism. The total modelled anchor length is $h_{eff} + t_u = 111.8$ mm.

The anchor, the washer, the nut and the plate are fully connected to one another and behave as a **monolithic body**. The radial clearance $t_{cl} = 0.5$ mm between the shaft and the plate hole is retained inside that body, so the assembly can develop **self contact** at the shaft/hole interface. Figure 1c gives that clearance together with the two surface pairs that carry contact: the shaft against the wall of the plate hole inside the monolithic steel part, and the plate underside against the concrete surface. The realisation of these connections is given in Section 1.2.2.

The assumed loading and the model are symmetric regarding the global z coordinate, thus only half of the model needs to be simulated (see also Figure 2). The retained half is $z \leq 0$, so the modelled concrete block measures $W_c \times H_c \times Z_c = 500 \times 240 \times 250$ mm and the modelled plate $b_p \times w_p \times t_p = 100 \times 200 \times 20$ mm.

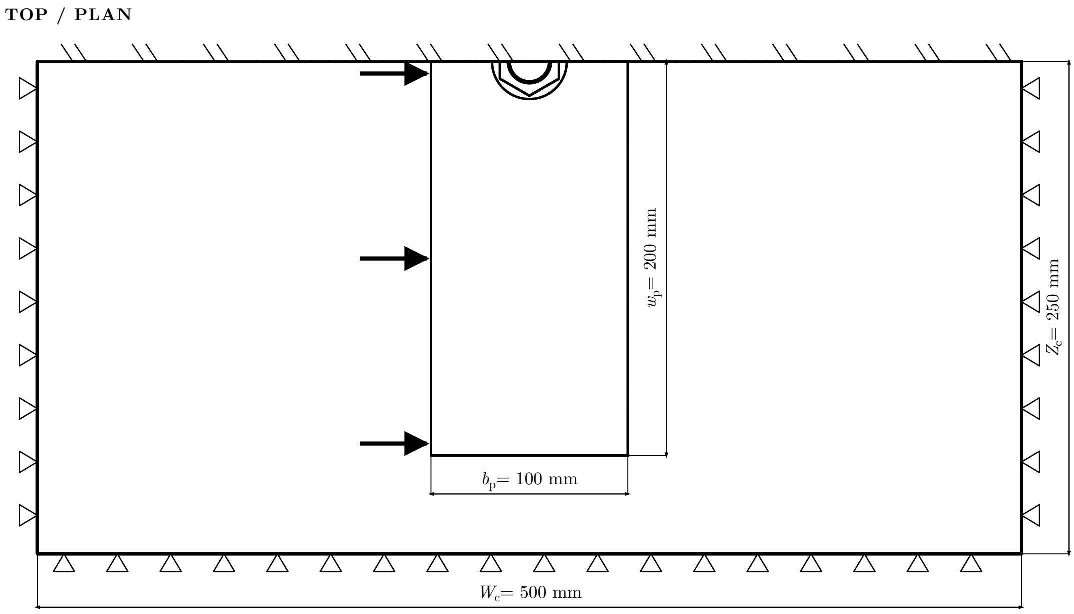


Figure 2: Top view of the model including the plane of symmetry $z = 0$, across which the model and the loading are symmetric. The prescribed displacement $u_x = 5$ mm is distributed over the 51 nodes of the plate edge set `plate_left_loading`.

The component dimensions are summarized in the next table:

1.1.2 Boundary conditions and loading

A uniform horizontal displacement boundary condition is applied on the left part of the steel plate at $t_p/3$ from the concrete surface.

Parameter	Value	Description
d_{a}	20 mm	Anchor diameter
h_{eff}	70 mm	Effective embedment depth; also the borehole depth
t_{m}	1 mm	Mortar thickness
d_{h}	22 mm	Borehole diameter, $d_{\mathrm{a}} + 2t_{\mathrm{m}}$
W_{c}	500 mm	Slab length and width
H_{c}	240 mm	Slab height, $> 3h_{\mathrm{eff}}$
Z_{c}	250 mm	Modelled slab half-width, $W_{\mathrm{c}}/2$
b_{p}	100 mm	Plate extent along x, the loading direction
l_{p}	400 mm	Plate extent along z
w_{p}	200 mm	Modelled plate half-length, $l_{\mathrm{p}}/2$
t_{p}	20 mm	Plate thickness
d_{cl}	21 mm	Plate hole and washer bore, $d_{\mathrm{a}} + 1$
t_{cl}	0.5 mm	Radial clearance, shaft to plate hole, $(d_{\mathrm{cl}} - d_{\mathrm{a}})/2$
d_{w}	38 mm	Washer outer diameter, $1.9d_{\mathrm{a}}$
t_{w}	3.8 mm	Washer thickness, $0.19d_{\mathrm{a}}$
d_{n}	30 mm	Nut width across flats, $1.5d_{\mathrm{a}}$
t_{n}	18 mm	Nut height, $0.9d_{\mathrm{a}}$
t_{u}	41.8 mm	Anchor protrusion above the concrete, $t_{\mathrm{p}} + t_{\mathrm{w}} + t_{\mathrm{n}}$
$t_{\mathrm{p}}/3$	6.667 mm	Height of the applied displacement above the concrete surface
u_{x}	5 mm	Prescribed horizontal displacement

Table 4: Geometrical parameters of the bonded anchor model with explicit washer–nut fixture.

1.2 Example FE model

The model is constructed and discretized with the help of Cubit [1] and then exported to Abaqus [2] format.

1.2.1 Discretization

20 node quadratic brick finite elements with reduced integration for the concrete (`C3D20R`), linear 8 noded brick finite elements with incompatible modes for the steel anchor, plate, washer and nut (`C3D8I`) and linear 8 noded cohesive elements (`COH3D8`) for the mortar are used. The resulting mesh is summarised in Table 5.

Part	Region	Elements	Element type
`concrete`	`concrete`	33 332	`C3D20R` (144 447 nodes)
`steel`	`anchor`	1 938	`C3D8I`
	`plate`	7 272	`C3D8I`
	`washer`	108	`C3D8I`
	`nut`	245	`C3D8I`
	`mortar`	392	`COH3D8`
13 255 nodes in the `steel` part in total			

Table 5: Mesh inventory of the half model. Target element sizes: 4.0 mm for the anchor and plate, $d_{\mathrm{a}}/8 = 2.5$ mm for the washer, the nut and the mortar sleeve, and 9.0 mm for the concrete, refined to 3.0 mm in the region surrounding the borehole.

The mortar is modeled with the use of the traction–separation bond law available in Abaqus, declared as

```
*COHESIVE SECTION, elset=MORTAR, material=stick_slip,
```

```
response=TRACTION SEPARATION, ORIENTATION=MORTAR_CYL,
        STACK DIRECTION=ORIENTATION
```

where MORTAR_CYL is a cylindrical system whose axis is the anchor axis: the first stiffness of the law acts along the borehole radius, the two remaining ones along the sleeve's circumferential and axial directions. The bond response is linear elastic in traction–separation. Table 6 contains k_{nm}, the mortar stiffness ([MPa/mm]) in the normal direction, k_{tm}, the mortar stiffness ([MPa/mm]) in the tangential direction (same for both tangential directions), and the mortar density ρ_{m}.

Parameter	**Value**	**Units**
k_{nm}	5400	MPa/mm
k_{tm}	2700	MPa/mm
ρ_{m}	2×10^{-6}	kg/mm^3

Table 6: Mortar properties.

1.2.2 Interaction between the structural components

The model interactions are five tie constraints (Table 7) and two contact pairs (Table 8). Surfaces are listed without their instance prefix: concrete_to_mortar and concrete_top belong to instance concrete-1, the remaining ones to instance steel-1.

Tie constraint	**Secondary surface**	**Main surface**
NUT_TO_ANCHOR	anchor_to_nut	nut_to_anchor
NUT_TO_WASHER	nut_to_washer	washer_to_nut
WASHER_TO_PLATE	washer_bottom	plate_top
ANCHOR_TO_MORTAR	mortar_to_anchor	anchor_to_mortar
MORTAR_TO_CONCRETE	concrete_to_mortar	mortar_to_concrete

Table 7: Tie constraints (*TIE) of the explicit-fixture model. All five are TYPE=SURFACE TO SURFACE with ADJUST=NO; NUT_TO_WASHER additionally carries POSITION TOLERANCE=0.6.

Both contact pairs — plate on concrete, and shaft in the plate hole, the latter a self contact between two surfaces of instance steel-1 — use a hard, frictionless interface law, declared as

```
*Surface Behavior, DIRECT, pressure-overclosure=HARD
*Contact Pair, interaction=..., type=SURFACE TO SURFACE
```

i.e. surface-to-surface discretisation, finite sliding, and DIRECT (Lagrange multiplier) enforcement of the hard pressure–overclosure law.

Interaction	**Secondary surface**	**Main surface**
"steel concrete"	plate_bottom	concrete_top
"steel steel"	anchor_to_plate	plate_to_anchor

Table 8: Contact pairs of the explicit-fixture model.

1.2.3 Boundary conditions in the input deck

The model-specific node sets, locations and constrained degrees of freedom are listed in Table 9. Apply them together with Table 3.

The model is cut on $z = 0$: the shear of the full specimen is twice the reaction summed over plate_left_loading.

Node set	Location	DOFs	Condition
`steel-1.z_symm`	$z = 0$; anchor, plate, washer, nut and mortar	3	`zSYMM`: $u_3 = 0$
`steel-1.plate_left_loading`	$x = -50$, $y = t_\mathrm{p}/3 = 6.667$, $-200 \le z \le 0$	1	$u_1 = 0$ in the base state; $u_1 = u_\mathrm{x} = +5$ mm prescribed in step `loading`

Table 9: Model-specific boundary conditions of the explicit-fixture model.

1.2.4 Solution procedure

An implicit dynamic analysis is performed using the Backward Euler solver of Abaqus, in one step with adaptive time incrementation:

```
*Step, name=loading, nlgeom=no, inc=100000
*Dynamic, APPLICATION=QUASI-STATIC
```

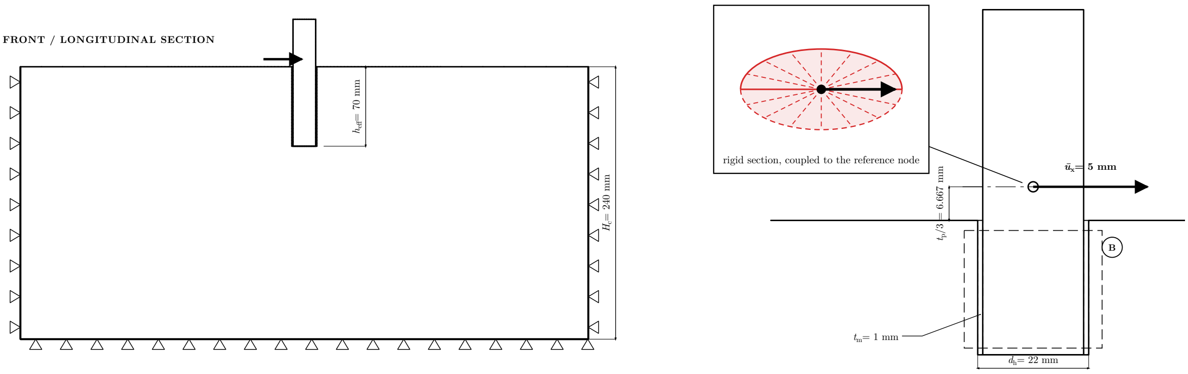
$$\Delta t_0 = 10^{-4}, \quad T = 1, \quad \Delta t_\mathrm{min} = 10^{-8}, \quad \Delta t_\mathrm{max} = 10^{-2}$$

Loading is displacement-controlled: $u_\mathrm{x} = 5$ mm is reached at the end of the step through `*Boundary, type=displacement` on `plate_left_loading`. Geometric nonlinearity is disabled (`nlgeom=no`) and restart data are written every increment.

2 Model 2: Bare anchor with kinematic constraint

2.1 Model setup

This document describes the **bare-anchor** variant of the model family. The model comprises the concrete slab, the mortar sleeve and the anchor, and the shear is applied directly to the anchor shaft. The complete shaft cross-section at the height $t_\mathrm{p}/3 = 6.667$ mm above the concrete surface is constrained to move as a rigid diaphragm in the loading plane, and the displacement is prescribed on the reference node of that constraint.



(a) (b)

Figure 3: Bare bonded-anchor model. a Section on the symmetry plane $z = 0$, together with the boundary conditions of the concrete slab; the dot marks the reference node of the rigid cross-section, on the anchor axis at $t_\mathrm{p}/3 = 6.667$ mm above the concrete surface, where the horizontal displacement is prescribed. b Detail near the anchor: mortar thickness t_m, borehole diameter $d_\mathrm{h} = d_\mathrm{a} + 2t_\mathrm{m}$, and the plane $t_\mathrm{p}/3$ on which the displacement is prescribed. The inset shows the complete shaft cross-section on that plane: the reference node lies at its centre on the anchor axis, and the spokes stand for the kinematic coupling that ties every node of the section to it.

2.1.1 Geometry

The model consists of a cylindrical steel anchor with diameter $d_\mathrm{a} = 20$ mm, embedded till depth $h_\mathrm{eff} = 70$ mm, in an orthogonal parallelepiped concrete slab with dimensions (length:

$W_c = 500$ mm, width: $W_c = 500$ mm, height: $H_c = 240$ mm $> 3h_{\text{eff}}$). A cylindrical hole is present inside the concrete slab with diameter $d_h = d_a + 2t_m = 22$ mm, where $t_m = 1$ mm is the mortar thickness. The anchor is placed inside the hole so that their axes along the height match. The mortar region is modeled as a hollow cylinder sleeve with height h_{eff} and constant thickness t_m between the anchor and the surrounding concrete.

The steel part of the model contains the anchor and the cohesive mortar sleeve. The shaft protrudes $t_u = 41.8$ mm above the concrete surface, and the displacement is applied on the plane $t_p/3 = 6.667$ mm above it, where $t_p = 20$ mm is the plate thickness of the explicit-fixture variant.

The assumed loading and the model are symmetric regarding the global z coordinate, thus only half of the model needs to be simulated (see also Figure 4). The retained half is $z \leq 0$, so the modelled concrete block measures $W_c \times H_c \times Z_c = 500 \times 240 \times 250$ mm.

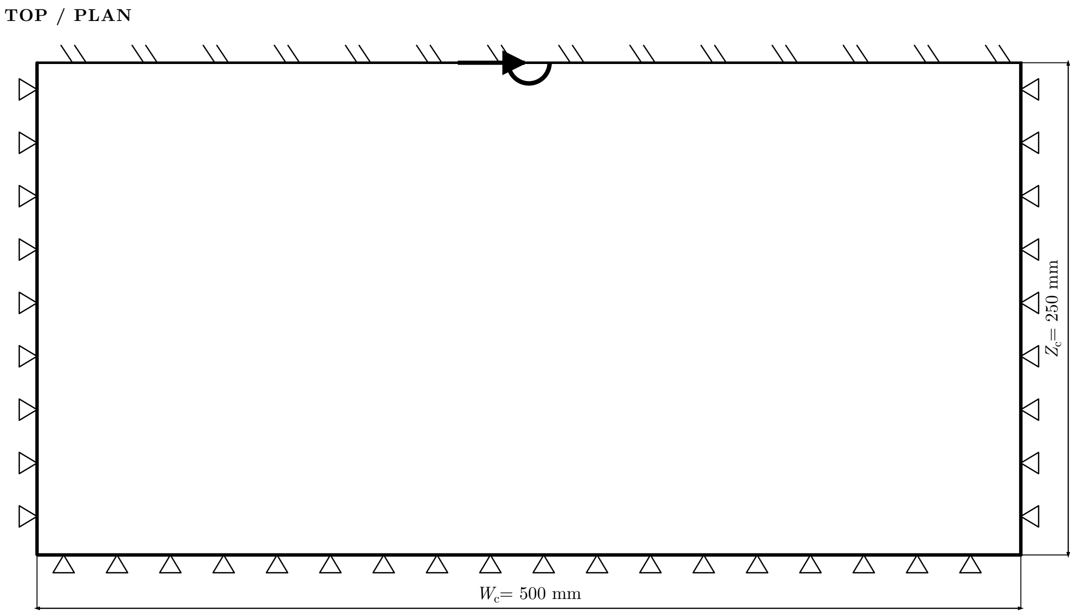


Figure 4: Top view of the model including the plane of symmetry $z = 0$, across which the model and the loading are symmetric. The prescribed displacement $u_x = 5$ mm is applied at the single reference node of the rigid cross-section, which lies on the anchor axis.

The component dimensions are summarized in the next table:

Parameter	Value	Description
d_a	20 mm	Anchor diameter
h_{eff}	70 mm	Effective embedment depth; also the borehole depth
t_m	1 mm	Mortar thickness
d_h	22 mm	Borehole diameter, $d_a + 2t_m$
W_c	500 mm	Slab length and width
H_c	240 mm	Slab height, $> 3h_{\text{eff}}$
Z_c	250 mm	Modelled slab half-width, $W_c/2$
t_u	41.8 mm	Free length of the shaft above the concrete
$t_p/3$	6.667 mm	Height of the loaded cross-section above the concrete surface, with $t_p = 20$ mm
u_x	5 mm	Prescribed horizontal displacement at the reference node

Table 10: Geometrical parameters of the bare bonded-anchor model.

2.1.2 Boundary conditions and loading

A horizontal displacement boundary condition is applied to the shaft cross-section at $t_\mathrm{p}/3$ from the concrete surface, which remains plane and rigid in the loading plane.

2.2 Example FE model

The model is constructed and discretized with the help of Cubit [1] and then exported to Abaqus [2] format.

2.2.1 Discretization

20 node quadratic brick finite elements with reduced integration for the concrete (`C3D20R`), linear 8 noded brick finite elements with incompatible modes for the steel anchor (`C3D8I`) and linear 8 noded cohesive elements (`COH3D8`) for the mortar are used. The resulting mesh is summarised in Table 11.

Part	**Region**	**Elements**	**Element type**
`concrete`	`concrete`	33 332	`C3D20R` (144 447 nodes)
`steel`	`anchor`	1 938	`C3D8I`
	`mortar`	392	`COH3D8`
	3 522 nodes in the `steel` part in total		

Table 11: Mesh inventory of the half model. Target element sizes: 4.0 mm for the anchor, $d_\mathrm{a}/8 = 2.5$ mm for the mortar sleeve, and 9.0 mm for the concrete, refined to 3.0 mm in the region surrounding the borehole.

The mortar is modeled with the use of the traction–separation bond law available in Abaqus, declared as

```
*COHESIVE SECTION, elset=MORTAR, material=stick_slip,
response=TRACTION SEPARATION, ORIENTATION=MORTAR_CYL,
        STACK DIRECTION=ORIENTATION
```

where `MORTAR_CYL` is a cylindrical system whose axis is the anchor axis: the first stiffness of the law acts along the borehole radius, the two remaining ones along the sleeve's circumferential and axial directions. The bond response is linear elastic in traction–separation. Table 12 contains k_nm, the mortar stiffness ([MPa/mm]) in the normal direction, k_tm, the mortar stiffness ([MPa/mm]) in the tangential direction (same for both tangential directions), and the mortar density ρ_m.

Parameter	**Value**	**Units**
k_nm	5400	MPa/mm
k_tm	2700	MPa/mm
ρ_m	2×10^{-6}	kg/mm^3

Table 12: Mortar properties.

2.2.2 Interaction between the structural components

The model interactions are the two tie constraints of Table 13 and the coupling constraint of Section 2.2.3. Surfaces are listed without their instance prefix: `concrete_to_mortar` belongs to instance `concrete-1`, the remaining three to instance `steel-1`.

Tie constraint	Secondary surface	Main surface
ANCHOR_TO_MORTAR	mortar_to_anchor	anchor_to_mortar
MORTAR_TO_CONCRETE	concrete_to_mortar	mortar_to_concrete

Table 13: Tie constraints (*TIE) of the bare-anchor model. Both are TYPE=SURFACE TO SURFACE with ADJUST=NO.

2.2.3 Load introduction: the rigid cross-section

A reference node SECRP (node 999002) is placed on the anchor axis at $(0,\ t_\mathrm{p}/3,\ 0) = (0,\ 6.667,\ 0)$. The 51 nodes of the complete shaft cross-section on that plane — perimeter and interior — are collected in the node set anchor_load_section and wrapped as a node-based surface, and the two are joined by a kinematic coupling:

```
*Surface, type=NODE, name=sec_nodes
steel-1.anchor_load_section, 1.
*Coupling, constraint name=SEC_RIGID, ref node=SECRP, surface=sec_nodes
*Kinematic
1, 2
```

Translational degrees of freedom 1 and 2 are coupled: the cross-section is rigid in the loading plane x–y.

2.2.4 Boundary conditions in the input deck

The model-specific node sets, locations and constrained degrees of freedom are listed in Table 14. Apply them together with Table 3.

Node set	Location	DOFs	Condition
steel-1.z_symm	$z = 0$; anchor and mortar	3	zSYMM: $u_3 = 0$
SECRP	node 999002 at (0, 6.667, 0)	3,4,5	$u_3 = \mathrm{UR}_1 = \mathrm{UR}_2 = 0$, the half-model symmetry triple
SECRP	as above	1	$u_1 = u_\mathrm{x} = +5$ mm prescribed in step loading

Table 14: Model-specific boundary conditions of the bare-anchor model.

The model is cut on $z = 0$: the shear of the full specimen is $2\,|\mathrm{RF}_1|$ at SECRP.

2.2.5 Solution procedure

An implicit dynamic analysis is performed using the Backward Euler solver of Abaqus, in one step with adaptive time incrementation:

```
*Step, name=loading, nlgeom=no, inc=100000
*Dynamic, APPLICATION=QUASI-STATIC
```

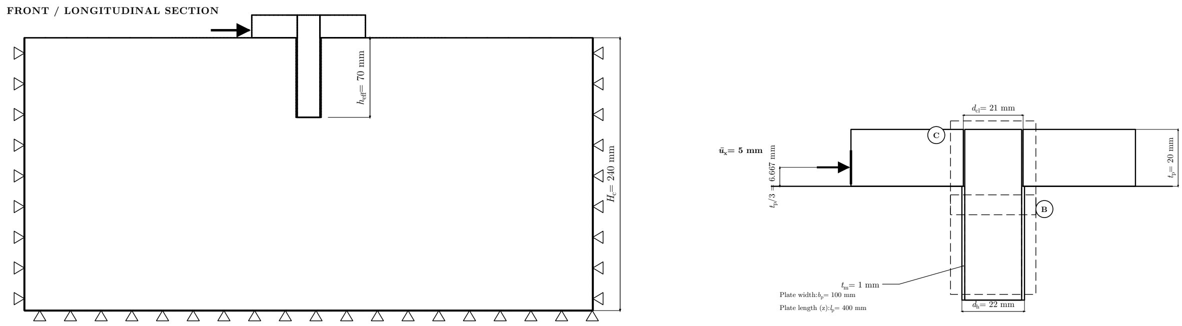
$$\Delta t_0 = 10^{-4}, \quad T = 1, \quad \Delta t_\mathrm{min} = 10^{-8}, \quad \Delta t_\mathrm{max} = 10^{-2}$$

Loading is displacement-controlled: $u_\mathrm{x} = 5$ mm is prescribed at SECRP through *Boundary, type=displacement and reached at the end of the step. Geometric nonlinearity is disabled (nlgeom=no) and restart data are written every increment.

3 Model 3: Merged anchor and plate

3.1 Model setup

This document describes the **merged-plate** variant of the model family: the anchor is extended above the concrete slab by the thickness of the steel plate, placed in the centre of a plate hole cut at exactly the anchor diameter, and the two components are then merged together so that they behave as one body for the subsequent discretization. The merge is performed on the mesh, so the shaft and the plate share nodes at their common surface.



(a) (b)

Figure 5: Bonded anchor model with the merged anchor–plate connection. a Section on the symmetry plane $z = 0$, together with the boundary conditions of the concrete slab; the horizontal displacement u_{x} is introduced on the negative-x edge of the steel plate, at the height $t_{\mathrm{p}}/3 = 6.667$ mm above the concrete surface, and the shaft terminates flush with the plate top at $y = t_{\mathrm{p}}$. b Detail near the anchor: mortar thickness t_{m}, borehole diameter $d_{\mathrm{h}} = d_{\mathrm{a}} + 2t_{\mathrm{m}}$ and plate thickness t_{p}; the plate hole is coincident with the shaft.

3.1.1 Geometry

The model consists of a cylindrical steel anchor with diameter $d_{\mathrm{a}} = 20$ mm, embedded till depth $h_{\mathrm{eff}} = 70$ mm, in an orthogonal parallelepiped concrete slab with dimensions (length: $W_{\mathrm{c}} = 500$ mm, width: $W_{\mathrm{c}} = 500$ mm, height: $H_{\mathrm{c}} = 240$ mm $> 3h_{\mathrm{eff}}$). A cylindrical hole is present inside the concrete slab with diameter $d_{\mathrm{h}} = d_{\mathrm{a}} + 2t_{\mathrm{m}} = 22$ mm, where $t_{\mathrm{m}} = 1$ mm is the mortar thickness. The anchor is placed inside the hole so that their axes along the height match. The mortar region is modeled as a hollow cylinder sleeve with height h_{eff} and constant thickness t_{m} between the anchor and the surrounding concrete.

The anchor is connected to a steel plate on top with dimensions (extent along x: $b_{\mathrm{p}} = 100$ mm, extent along z: $l_{\mathrm{p}} = 400$ mm, height: $t_{\mathrm{p}} = 20$ mm). The plate contains a hole in the middle with diameter matching the diameter of the anchor, $d_{\mathrm{a}} = 20$ mm, through its thickness. The connection of the anchor to the plate is as follows: first, the anchor is extended above the concrete slab by a length equal to the thickness of the steel plate, $t_{\mathrm{u}} = t_{\mathrm{p}} = 20$ mm, so that the shaft top is flush with the plate top; second, the anchor extension is placed in the centre of the steel plate hole. The two components (anchor, plate) are then merged together, behaving as one body for the subsequent discretizations of the model. The anchor and the plate are kept as separate element sets.

The assumed loading and the model are symmetric regarding the global z coordinate, thus only half of the model needs to be simulated (see also Figure 6). The retained half is $z \leq 0$, so the modelled concrete block measures $W_{\mathrm{c}} \times H_{\mathrm{c}} \times Z_{\mathrm{c}} = 500 \times 240 \times 250$ mm and the modelled plate $b_{\mathrm{p}} \times w_{\mathrm{p}} \times t_{\mathrm{p}} = 100 \times 200 \times 20$ mm.

The component dimensions are summarized in the next table:

3.1.2 Boundary conditions and loading

A uniform horizontal displacement boundary condition is applied on the left part of the steel plate at $t_{\mathrm{p}}/3$ from the concrete surface.

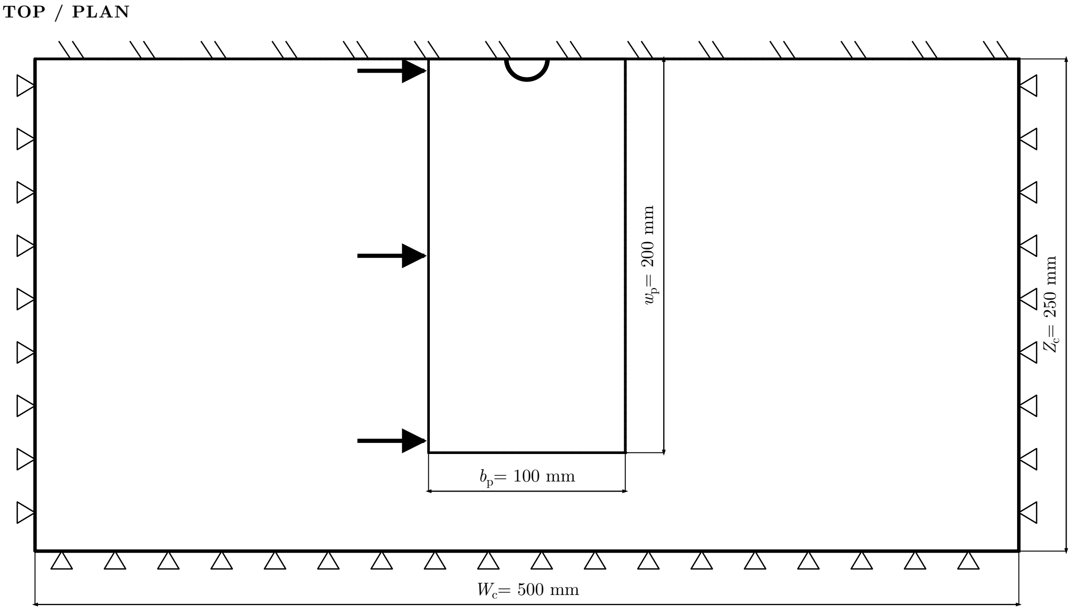


Figure 6: Top view of the model including the plane of symmetry $z = 0$, across which the model and the loading are symmetric. The prescribed displacement $u_x = 5$ mm is distributed over the 51 nodes of the plate edge set `plate_left_loading`.

Parameter	**Value**	**Description**
d_a	20 mm	Anchor diameter; also the plate hole diameter (zero clearance)
h_{eff}	70 mm	Effective embedment depth; also the borehole depth
t_m	1 mm	Mortar thickness
d_h	22 mm	Borehole diameter, $d_a + 2t_m$
W_c	500 mm	Slab length and width
H_c	240 mm	Slab height, $> 3h_{eff}$
Z_c	250 mm	Modelled slab half-width, $W_c/2$
b_p	100 mm	Plate extent along x, the loading direction
l_p	400 mm	Plate extent along z
w_p	200 mm	Modelled plate half-length, $l_p/2$
t_p	20 mm	Plate thickness
t_u	20 mm	Anchor protrusion above the concrete, $= t_p$, flush with the plate top
$t_p/3$	6.667 mm	Height of the applied displacement above the concrete surface
u_x	5 mm	Prescribed horizontal displacement

Table 15: Geometrical parameters of the bonded anchor model with merged anchor–plate connection.

3.2 Example FE model

The model is constructed and discretized with the help of Cubit [1] and then exported to Abaqus [2] format. The merge of the anchor and the plate is performed in Cubit before the mesh is written.

3.2.1 Discretization

20 node quadratic brick finite elements with reduced integration for the concrete (`C3D20R`), linear 8 noded brick finite elements with incompatible modes for the steel plate and anchor (`C3D8I`) and linear 8 noded cohesive elements (`COH3D8`) for the mortar are used. The resulting mesh is summarised in Table 16.

Part	**Region**	**Elements**	**Element type**
`concrete`	`concrete`	33 332	`C3D20R` (144 447 nodes)
`steel`	`anchor`	1 026	`C3D8I`
	`plate`	11 808	`C3D8I`
	`mortar`	392	`COH3D8`
	16 128 nodes in the `steel` part in total		

Table 16: Mesh inventory of the half model. Target element sizes: 4.0 mm for the merged anchor–plate body, $d_{\mathrm{a}}/8 = 2.5$ mm for the mortar sleeve, and 9.0 mm for the concrete, refined to 3.0 mm in the region surrounding the borehole.

The mortar is modeled with the use of the traction–separation bond law available in Abaqus, declared as

```
*COHESIVE SECTION, elset=MORTAR, material=stick_slip,
response=TRACTION SEPARATION, ORIENTATION=MORTAR_CYL,
STACK DIRECTION=ORIENTATION
```

where `MORTAR_CYL` is a cylindrical system whose axis is the anchor axis: the first stiffness of the law acts along the borehole radius, the two remaining ones along the sleeve's circumferential and axial directions. The bond response is linear elastic in traction–separation. Table 17 contains k_{nm}, the mortar stiffness ([MPa/mm]) in the normal direction, k_{tm}, the mortar stiffness ([MPa/mm]) in the tangential direction (same for both tangential directions), and the mortar density ρ_{m}.

Parameter	**Value**	**Units**
k_{nm}	5400	MPa/mm
k_{tm}	2700	MPa/mm
ρ_{m}	2×10^{-6}	kg/mm^3

Table 17: Mortar properties.

3.2.2 Interaction between the structural components

The model interactions are the two tie constraints of Table 18, the contact pair of Table 19, and the merged shaft/plate interface described at the end of this section. Surfaces are listed without their instance prefix: `concrete_to_mortar` and `concrete_top` belong to instance `concrete-1`, the remaining ones to instance `steel-1`.

The contact pair between the plate underside and the concrete surface uses a hard, frictionless interface law, declared as

```
*Surface Behavior, DIRECT, pressure-overclosure=HARD
*Contact Pair, interaction="steel concrete", type=SURFACE TO SURFACE
```

Tie constraint	Secondary surface	Main surface
ANCHOR_TO_MORTAR	mortar_to_anchor	anchor_to_mortar
MORTAR_TO_CONCRETE	concrete_to_mortar	mortar_to_concrete

Table 18: Tie constraints (*TIE) of the merged-plate model. Both are TYPE=SURFACE TO SURFACE with ADJUST=NO.

i.e. surface-to-surface discretisation, finite sliding, and DIRECT (Lagrange multiplier) enforcement of the hard pressure–overclosure law.

Interaction	Secondary surface	Main surface
"steel concrete"	plate_bottom	concrete_top

Table 19: Contact pairs of the merged-plate model.

The plate hole is cut at the anchor radius and the two bodies are merged before meshing: 150 nodes are common to the ANCHOR and PLATE element sets, lying on the cylinder $r = d_a/2 = 10$ mm over $0 \le y \le t_p$.

3.2.3 Boundary conditions in the input deck

The model-specific node sets, locations and constrained degrees of freedom are listed in Table 20. Apply them together with Table 3.

Node set	Location	DOFs	Condition
steel-1.z_symm	$z = 0$; merged anchor–plate body and mortar	3	zSYMM: $u_3 = 0$
steel-1.plate_left_loading	$x = -50$, $y = t_p/3 = 6.667$, $-200 \le z \le 0$	1	$u_1 = 0$ in the base state; $u_1 = u_x = +5$ mm prescribed in step loading

Table 20: Model-specific boundary conditions of the merged-plate model.

The model is cut on $z = 0$: the shear of the full specimen is twice the reaction summed over plate_left_loading.

3.2.4 Solution procedure

An implicit dynamic analysis is performed using the Backward Euler solver of Abaqus, in one step with adaptive time incrementation:

```
*Step, name=loading, nlgeom=no, inc=100000
*Dynamic, APPLICATION=QUASI-STATIC
```

$$\Delta t_0 = 10^{-4}, \quad T = 1, \quad \Delta t_{\min} = 10^{-8}, \quad \Delta t_{\max} = 10^{-2}$$

Loading is displacement-controlled: $u_x = 5$ mm is reached at the end of the step through *Boundary, type=displacement on plate_left_loading. Geometric nonlinearity is disabled (nlgeom=no) and restart data are written every increment.

References

[1] Coreform LLC. *Coreform Cubit*, 2025. Computer software.

[2] Dassault Systèmes SIMULIA. *Abaqus 2024 Documentation*. Dassault Systèmes, 2024. Finite element analysis software.